Logistic Regression

Supervised Machine Learning
Classification Algorithm

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Course Topics

- ✔ What is Logistic Regression?
- ✔ How the algorithm works
- ✔ Logistic Regression assumptions
- ✔ Pros and Cons
- ✔ Applications
- ✔ Modeling Logistic Regression with Python

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What is Logistic Regression?

- Logistic Regression is a classification algorithm that can predict a categorical variable.
- The dependent variable is a binary variable that contains data coded as 1 (Yes/True..) or 0 (No/False..).
- The model arrives at the prediction by computing the probability of a class, say Class 1 to be the correct label.

Study Hours	Output (Pass/Fail)
2	Fail
3	Fail
4	Fail
5	Fail
6	Pass
7	Pass
8	Pass
9	Fail

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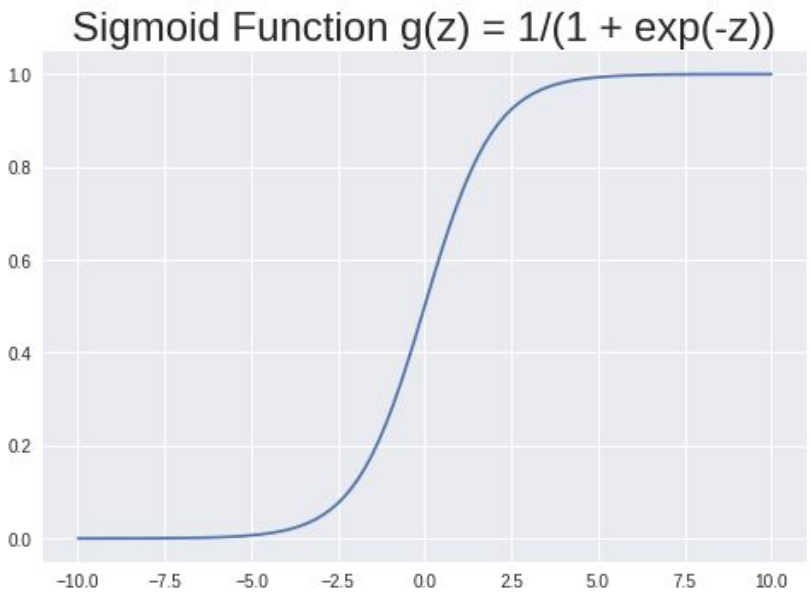
How Logistic Regression works?

$$h_a(x) = g(z) = \frac{1}{1 + e^{-z}}$$

$$h_a(x) = \frac{1}{1 + e^{-a'x}}$$

$$h_a(x) = g(a_0 + a_1 x_1 + a_2 x_2 + \cdots + a_n x_n)$$



Decision Boundaries

If $h_a(x) >= 0.5$ then we say $y = 1$
If $h_a(x) < 0.5$ then we say $y = 0$

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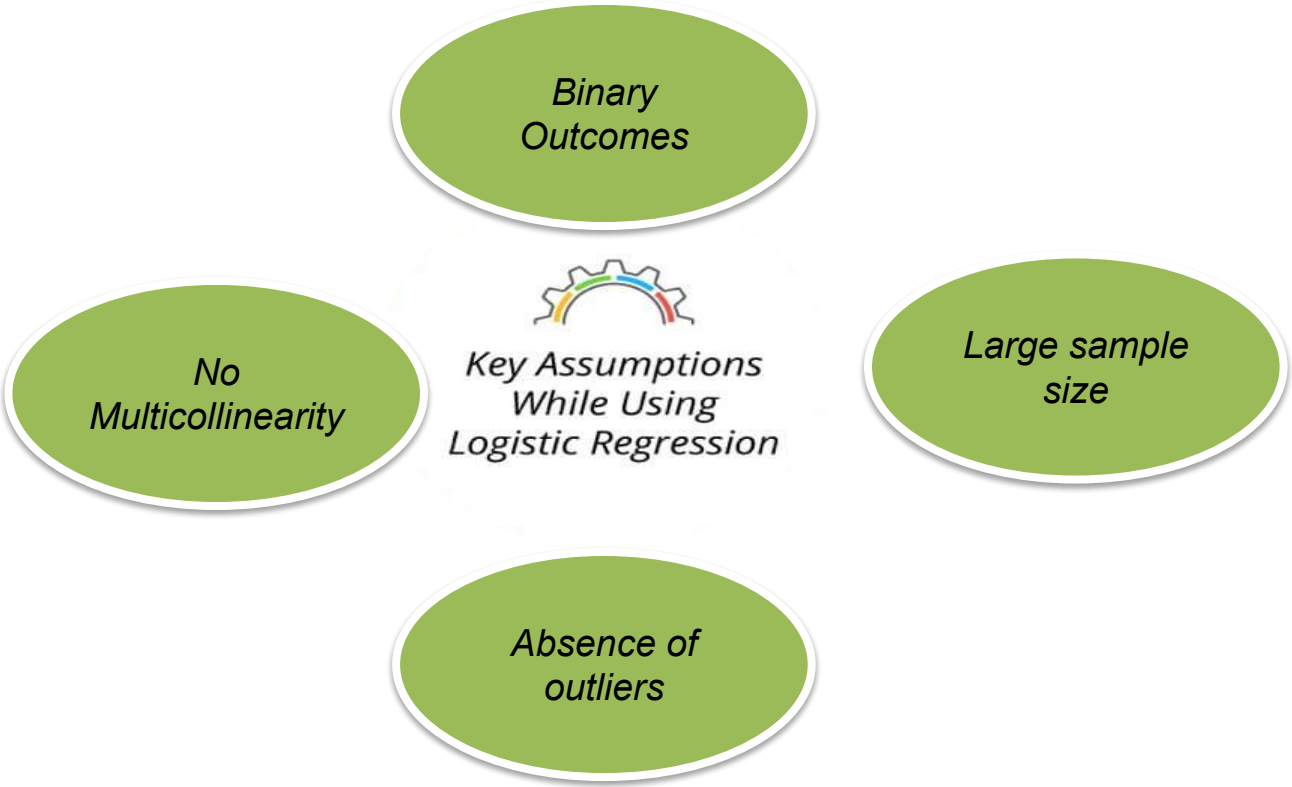
Assumptions of Logistic Regression



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Pros of Logistic Regression

Logistic regression works well for cases where the dataset is linearly separable.

Logistic regression is much easier to implement than other methods, especially in the context of machine learning.

Logistic regression provides useful insights.

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Cons of Logistic Regression

The major drawback is, it is vulnerable to overfitting.

It can not handle many categorical variables, which means it cannot be used on a heavy model.

If independent variables are not correlated with the target variable, then this technique does not work properly.

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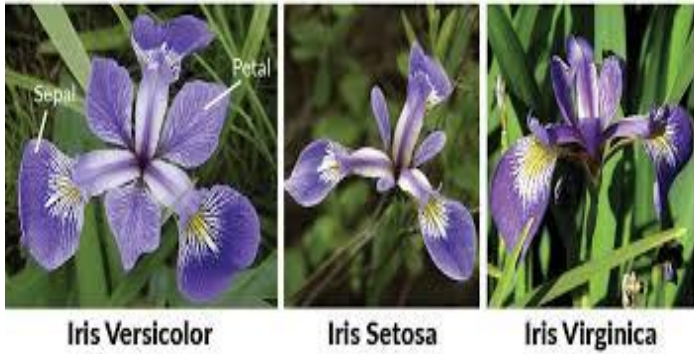
Applications of Logistic Regression

- Image Segmentation and Categorization
- Handwriting recognition
- Spam filtering -- spam or not spam
- Cell image -- cancer or normal
- Production line part scan -- good or defective



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Classification in action

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